

Table 4: **Effect of LoRA Rank on the Correlation Between Pathway Metrics and Generalization** (Commonsense reasoning domain). The results show that increasing the LoRA rank from 32 to 64 yields nearly identical correlations, confirming that the observed trends are robust to LoRA configuration. p -value: $p < 0.1$, $p < 0.05$, $p < 0.01$, $p > 0.1$, $p > 0.5$, $p > 0.8$. Preferred correlation: positive, negative.

Metric	LoRA Rank = 32		LoRA Rank = 64	
	Pearson	Spearman	Pearson	Spearman
Pathway Distance (pairwise)	-0.9821	-0.9747	-0.9927	-0.9487
Pathway Consistency (sample-wise)	0.9804	0.9747	0.9559	0.9487
Training Loss	0.6427	0.4104	0.6197	0.4617
Training Loss (moving average)	0.7931	0.6669	0.6871	0.2052
Validation Loss	-0.5752	-0.4617	-0.6631	-0.4104

Varying δ . Figure 11 reports the results when fixing $\varepsilon = 2.0$ and varying δ between 0.03 and 0.05. Similarly, the decreasing trend of edit distance after memorization persists, and the fitted trajectories show negligible variation across δ values.

This confirms that the proposed pathway-based observations are **robust to the choice of ε and δ** .

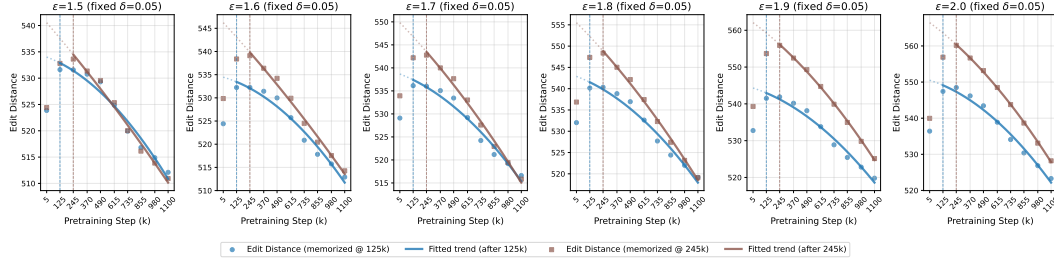


Figure 10: **Effect of the loss threshold ε on pathway dynamics.** The stability threshold is fixed at $\delta = 0.05$. All curves show a consistent post-memorization decrease in edit distance, indicating that the results are robust to ε .

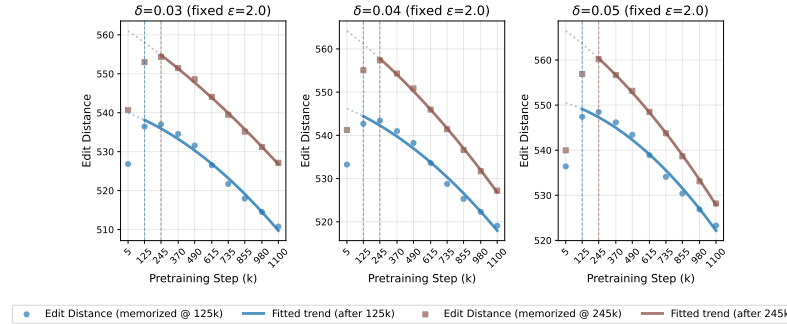


Figure 11: **Effect of the stability threshold δ on pathway dynamics.** The loss threshold is fixed at $\varepsilon = 2.0$. The same decreasing trend holds across different δ values, confirming robustness to this parameter.

E.4 Cross-Domain Analysis of Pathway Dynamics

To verify that the observed changes in pathway edit distance are due to the emergence of shared pathways within specific domains rather than uniform effects across all data, such as load balancing or regularization, we conduct an additional analysis focusing on the *cross-domain* samples. Using the same cumulative routing-weight threshold (0.7) as in the main experiments, we compute average pathway edit distances for samples drawn from two different domains across checkpoints—specifically, *math* $\times$ *code*—under identical configurations.

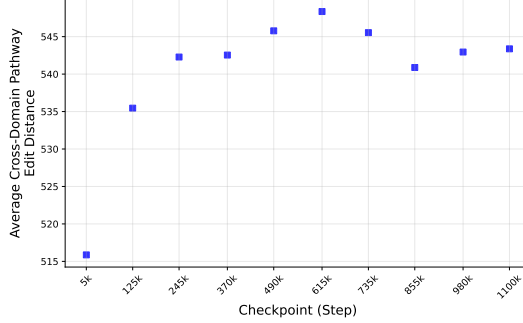


Figure 12: **Cross-domain pathway edit distance ($\text{math} \times \text{code}$)**. The figure shows the global trend of average pathway edit distance during pretraining when computed across two different domains. The cross-domain distances fluctuate irregularly, exhibiting both increases and decreases but no systematic downward trend. This contrasts with the localized decreases observed within domains (Fig. 4), indicating that pathway reorganization occurs locally among semantically related data rather than globally across domains.

As shown in Fig. 12, the cross-domain edit distance fluctuates throughout pretraining, showing irregular increases and decreases without any consistent downward trend. In contrast, Fig. 4 demonstrates that within-domain pathway distances decrease only for groups of samples that become memorized at different points in training, with each domain exhibiting its own localized transition rather than a global drop. These observations confirm that the structural changes we observe are **domain- and group-specific**, reflecting the **emergence of shared pathways among semantically related samples** rather than a global effect of routing behavior.

E.5 Relationship Between Routing Entropy and Edit Distance

We investigate whether changes in routing diversity could trivially explain the observed trends in pathway edit distance. Specifically, we test whether routing entropy—high when expert usage is balanced and low when a few experts dominate—can drive edit distance dynamics.

For each sample, routing entropy is computed *per layer* from the full routing-weight vector (no thresholding or selection) and averaged across layers. For each sample pair, we take the mean of the two samples’ entropies and compute their pathway edit distance using a cumulative routing-weight threshold of 0.8, which provides sufficient expert coverage for analysis.

We sample 141,530 pairs across checkpoints spanning both the *math* and *code* domains, including within-domain and cross-domain pairs. Figure 13 shows a weak correlation between average routing entropy and edit distance ($r = 0.24$, explaining about 6% of the variance) with no clear monotonic or directional trend. This indicates that neither more balanced routing (higher entropy) nor more concentrated routing (lower entropy) systematically reduces pathway distance. Therefore, the decrease in within-domain pathway distance observed after memorization is unlikely to result from overall changes in routing balance; instead, it reflects domain- and group-specific reorganization of computation.

E.6 Consistent Findings on another MoE (nanoMoE)

To further validate the generality of our findings beyond OLMoE, we conduct additional experiments on a 55M-parameter MoE model, **nanoMoE**¹. NanoMoE provides a small, fully trainable MoE architecture for which we can obtain complete pretraining checkpoints, making it suitable for analysis of routing dynamics.

Training setup. We train nanoMoE on $\sim 25\text{B}$ tokens of the OpenWebText dataset [7] for 50k iterations on two NVIDIA H100 GPUs and uniformly sample ten checkpoints for analysis. OpenWebText is a mixed-domain web corpus with a heavy emphasis on commonsense and general knowledge content. To approximate a domain setting, we construct a commonsense subset by filtering 10,000 randomly

¹<https://github.com/wolfecameron/nanoMoE>

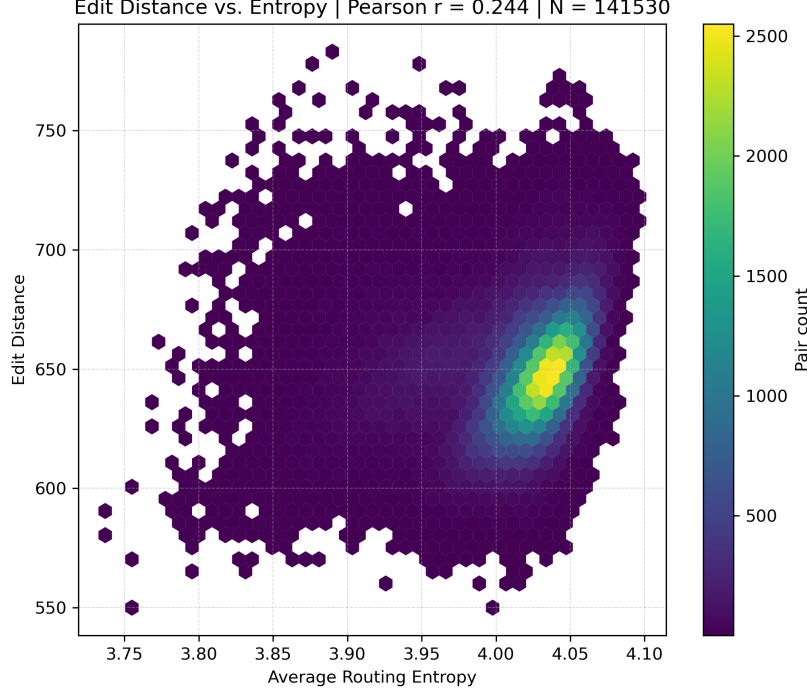


Figure 13: **Edit distance vs. routing entropy.** Each hexagon aggregates sample pairs ($N = 141,530$) drawn across checkpoints and across the *math* and *code* domains. Routing entropy is computed from the *full* routing-weight vectors (no thresholding); pathway edit distance uses pathways built with a cumulative cutoff of 0.8. The weak correlation ($r = 0.24$) indicates that routing balance does not systematically determine pathway similarity.

sampld training samples using Qwen3-8B [33] to select those semantically related to commonsense reasoning, yielding 7,857 samples in total. Following the same methodology as our main experiments, we identify memorized samples based on token-level loss stability and compute pathway dynamics for this subset across checkpoints.

Pathway-metric dynamics. Consistent with our large-scale results, we observe that after memorization, **pathway edit distance decreases while pathway consistency increases** as shown in Figure 14 and 15. These trends mirror the post-memorization reorganization we identify in OLMoE, demonstrating that even the smaller MoE model reorganizes routing toward more structured, shared computation after the loss has already converged.

Correlation with downstream performance. We further evaluate all ten checkpoints in a zero-shot setting on standard commonsense reasoning benchmarks (ARC-Easy/Challenge [4], HellaSwag [34], and OpenBookQA [20]). Zero-shot evaluation is justified because models pretrained on OpenWebText, including GPT-2 [26], have been shown to achieve strong zero-shot performance on commonsense benchmarks. As in the large-scale setting, both pathway metrics exhibit strong correlations with benchmark accuracy ($|r| \approx 0.8\text{--}1.0$), as reported in Table 5, substantially outperforming training loss or its moving average. This confirms that routing dynamics remain tightly coupled with downstream generalization.

The nanoMoE results reinforce our central claim: *post-memorization routing reorganization is not an artifact of model scale, data size, or specific MoE implementation*. Instead, it reflects an inherent property of MoE training dynamics. Combined with the large-scale OLMoE study, these findings establish pathway metrics as stable, architecture-intrinsic indicators of the memorization-to-generalization transition.